

Importance Sampling Video Light Sources with Codec Partitions

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Figure 1: Random, dense 2D CDF and codec sampler (Ours) at 33 ms against a converged reference; closeups show RMSE.

1 Introduction

We introduce an efficient technique for importance sampling emissive textures used in video-driven lighting for virtual production, AR/VR and visual effects, where an HDR video shown on an LED wall, a virtual screen or an environment map must illuminate synthetic geometry in real time. Monte Carlo renderers importance sample such a source in proportion to its radiance, but dense structures such as a 2D CDF must be rebuilt for every frame, at a high cost in time and memory bandwidth.

Compressed video formats such as VP9, HEVC and AV1 already represent each frame through coding trees refined where the signal varies [Han et al. 2021; Mukherjee et al. 2013; Sullivan et al. 2012], and the transform coefficients [Ahmed et al. 1974] of each block encode its mean (DC) and its low-frequency variation (AC). We repurpose this hierarchy as a sampling structure without additional spatial analysis: the mean luminance of each block turns the coding tree into an alias table that draws light samples in constant time, removing the dense CDF, lowering memory bandwidth and enabling real-time lighting from video. It extends a preliminary hierarchical-CDF version [Aguerre 2026].

2 Method and Results

We encode each light video with libvpx (VP9 Profile 2, 10-bit) at a speed setting that uses the variance-based partitioning rule, and export from the decoder each frame’s superblock partition (64×64 down to 8×8 pixels) and block coefficients. Each coded block becomes a leaf i with area A_i , mean luminance $\bar{L}_i$ taken from its DC term in linear light, and flux weight $w_i = A_i \bar{L}_i$. A Walker/Vose alias table over these weights is built in $O(n)$. A sample picks a leaf in $O(1)$ and a uniform point inside it, with density $\bar{L}_i / \sum_j w_j$, and an index map with one entry per 8×8 cell returns the same density for multiple importance sampling. At 16 bytes per leaf, the structure takes 0.12–0.8 MB and is built in under 1 ms per frame, versus about 10 ms (Full HD) and 33 MB (4K) for a dense CDF.

In a GPU path tracer calibrated to 33 ms per frame (RTX 4080 SUPER), the saved time buys more light samples, and the codec sampler has the lowest RMSE against a converged reference on Rover, Cinema and Fireplace Room, 10–40% below the dense CDF (Fig. 1 shows the first two). We also score densities q over every fifth frame by the relative variance of a one-sample flux estimate, $\chi^2 = \sum_x p^*(x)^2 / q(x) - 1$ with $p^* = L / \sum_x L$ (Table 1). The coding tree lowers χ^2 by 9–87× relative to uniform (Random) sampling and by 2.5–7.1× relative to 64×64 superblock means.

Table 1: χ^2 per frame (SB: superblock means; G8: 8×8 grid).

Sequence	Res.	Leaves	Uniform	SB	DC	+AC	G8
Rover	1080p	10.7k	0.636	0.186	0.071	0.045	0.044
Cinema	1080p	7.4k	0.958	0.087	0.035	0.018	0.011
Fireplace	4K	50.1k	8.153	0.664	0.094	0.037	0.070

3 Discussion

Other libvpx speed settings use rate-distortion search or reuse partitions, so their trees follow coding cost and motion. HEVC [Sullivan et al. 2012], AV1 [Han et al. 2021] and VVC [Bross et al. 2021] partitions, including binary, ternary and T-shaped splits, also yield rectangles the alias table handles unchanged; where a tree is not variance-driven, Parseval’s theorem gives a block’s variance from its AC energy, so leaves can be split or merged in the compressed domain.

The DC coefficient of a predicted block is the mean of its residual, so in inter frames it is a difference to the motion-compensated reference. As interpolation filters have unit DC gain, a DC image of the reference frame can track block means [Yeo and Liu 1995], with a drift that resets at keyframes; intra blocks use the reconstructed block. For HDR, the mean of PQ or HLG code values underestimates the linear mean of textured blocks. These errors add variance but no bias, since sampling and weighting share one density.

The first AC coefficients describe how brightness tilts across a block; a sloped density built from them inside each leaf keeps closed-form sampling and lowers χ^2 by a further 37–61% (Table 1), though it is untested in the renderer. Exposing these terms as decoder side data, which hardware decoders lack, would let a renderer sample an LED-wall feed directly.

References

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